

Traffic Manager · Google Ads API Tool Design Document

1. Company Overview

Company Name: Blossom Boost **Website:** <https://www.blossomboost.com.br/> **Industry:** Custom Software Development **Location:** Brazil

Blossom Boost is a custom software development company that builds tailored software solutions for clients. We use Google Ads to run paid advertising campaigns to acquire new clients for our services.

2. Tool Overview

Tool Name: Traffic Manager **Purpose:** Internal multi-platform ad campaign management tool **Users:** Internal marketing team only (2-5 employees) **Platforms Integrated:** Google Ads, Meta Ads, LinkedIn Ads

Traffic Manager is an internal tool that consolidates advertising data from multiple platforms (Google Ads, Meta Ads, LinkedIn Ads) into a single dashboard. It allows our marketing team to monitor campaign performance, manage campaign status, and generate cross-platform reports without switching between multiple ad platform interfaces.

3. Google Ads API Usage

3.1 Features Using the API

| Feature | API Resources Used | Read/Write | |-----|-----|-----| | Campaign data sync | GoogleAdsService (campaign, ad_group, ad_group_ad) | Read | | Performance metrics | GoogleAdsService (metrics, segments) | Read | | Campaign status management | CampaignService (mutate) | Write | | Budget adjustments | CampaignBudgetService (mutate) | Write | | Reporting | GoogleAdsService (search/searchStream) | Read |

3.2 API Operations

Read Operations (Primary Use · ~90% of API calls):

- Fetch campaign structures (campaigns, ad groups, ads)
- Pull daily performance metrics (impressions, clicks, spend, conversions, CTR, CPC, CPM, CPA, ROAS)
- Generate performance reports for date ranges

Write Operations (~10% of API calls):

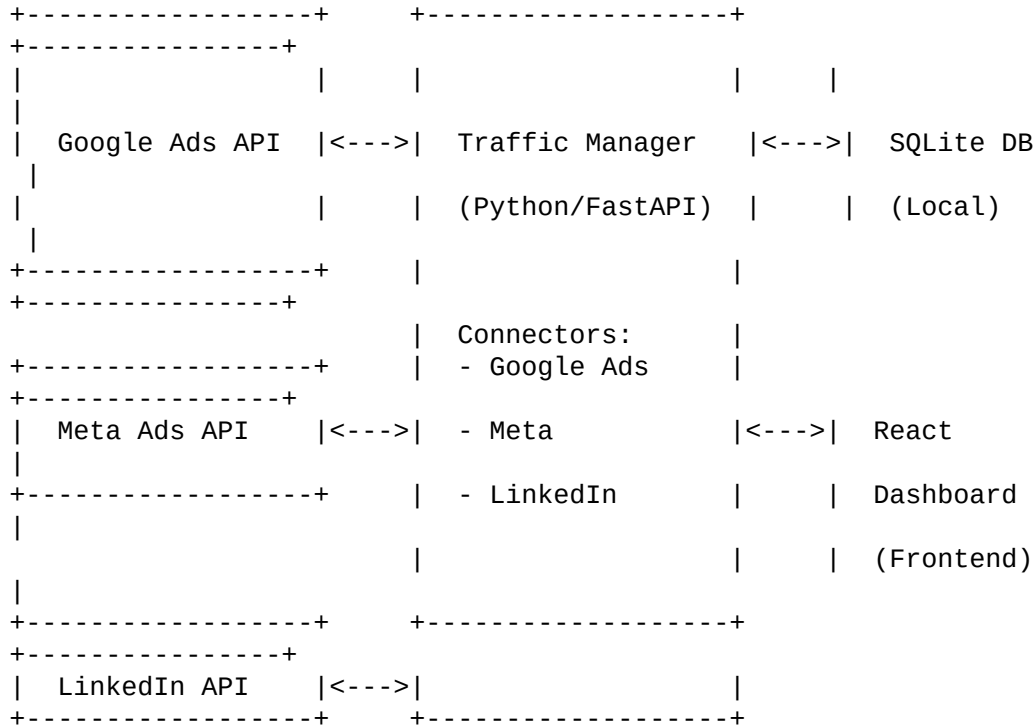
- Update campaign status (ENABLED, PAUSED)
- Update campaign budget amounts

3.3 API Call Frequency

- Automated metrics sync: Every 6 hours (4 times per day)
- Manual data refresh: On-demand by team members (estimated 5-10 times per day)
- Campaign updates: Occasional (estimated 2-5 per week)
- Total estimated daily API calls: Under 1,000

4. Technical Architecture

4.1 System Diagram



4.2 Technology Stack

- **Backend:** Python 3.12, FastAPI
- **Google Ads SDK:** google-ads Python client library (v29+)
- **Database:** SQLite (local storage for metrics history)
- **Frontend:** React (dashboard for data visualization)
- **Authentication:** OAuth 2.0 (Desktop application flow)

4.3 Data Flow

- 1 **Sync Process:** Scheduled job connects to Google Ads API via `google-ads` Python SDK
- 2 **Data Retrieval:** Uses `GoogleAdsService.SearchStream` to fetch campaign data and metrics
- 3 **Normalization:** Campaign data is normalized into a unified schema (shared with Meta and LinkedIn data)
- 4 **Storage:** Normalized data is stored in local SQLite database
- 5 **Display:** React dashboard reads from the API and displays consolidated metrics
- 6 **Actions:** Campaign status changes and budget updates are sent back to Google Ads API via `CampaignService.Mutate`

4.4 Authentication Flow

- OAuth 2.0 Desktop Application flow
- Refresh token stored securely in environment variables
- Access tokens refreshed automatically by the google-ads SDK
- Single Google Ads account (Customer ID: 743-337-0539) managed through MCC (207-592-5672)

5. Data Handling

5.1 Data Stored

- Campaign metadata (name, status, objective, budget)
- Ad group metadata (name, status, targeting)
- Ad metadata (name, status, creative info)
- Daily performance metrics (impressions, clicks, spend, conversions, revenue)
- Calculated metrics (CTR, CPC, CPM, CPA, ROAS)

5.2 Data Retention

- Performance metrics: Retained for 180 days for trend analysis
- Campaign metadata: Updated on each sync, current state only

5.3 Data Security

- All credentials stored in environment variables (never in code)
- Application runs on internal infrastructure only
- No data shared with third parties
- SQLite database stored locally with filesystem permissions

6. Compliance

- The tool is used exclusively by Blossom Boost internal employees
- No customer data from Google Ads end users is collected or stored
- The tool does not resell or redistribute Google Ads data
- All API usage complies with Google Ads API Terms and Conditions
- API contact email is monitored and kept up-to-date

7. Rate Limiting and Error Handling

- API calls are batched and throttled to stay within rate limits
- Exponential backoff retry logic for transient errors
- Failed syncs are logged with error details for debugging
- Metrics sync runs on a fixed schedule (every 6 hours) to avoid excessive API calls

8. Screenshots

The tool is currently in development (Phase 0). The dashboard will display:

- Campaign performance overview (spend, conversions, CPA, ROAS)
- Cross-platform comparison charts (Google Ads vs Meta vs LinkedIn)
- Campaign status management interface
- Automated alert notifications for performance anomalies